INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Probability Axioms & Distributions

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Probability Axioms & Distributions

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/probability/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6, quiz-7, quiz-8

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Probability P(E) = Number of Favourable Outcomes / Total Number of Elementary Outcomes.
- 0 <= P(E) <= 1; P(E) + P(not E) = 1.
- Addition Theorem: P(A or B) = P(A) + P(B) - P(A and B). For mutually exclusive events: P(A or B) = P(A) + P(B).
- Independent Events: P(A and B) = P(A) * P(B).
- Conditional Probability: P(A | B) = P(A and B) / P(B).

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Complementary rule for 'at least one': P(at least one success) = 1 - P(no successes).

- Two dice sum shortcuts: Sum 7 is most frequent (6/36 = 1/6); Sum 2 or 12 has 1/36.
- Pack of 52 cards: 4 suits of 13 cards; 12 face cards (Jack, Queen, King); 26 red, 26 black.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
In a horse racing competition, there were 18 numbered 1 to 18. The organizers assigned a probability of winning the race to each horse based on horses health and training the probability that horse one would win is 1/7, that 2 would win is 1/8, and that 3 would win is 1/7.Assuming that tie is impossible. Find the chance that one of these three will win the race?

Options:
- **(A)** `\frac {22}{392}`
- **(B)** `\frac {1}{396}`
- **(C)** `\frac {23}{56}`
- **(D)** `\frac {231}{561}`

Correct Answer: (C) `rac {23}{56}`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
CORRECT ANSWER IS\frac {23}{56}


HORSE 1: \frac {1}{7} WINNING PROBABILITY


HORSE 2: \frac {1}{8} WINNING PROBABILITY


HORSE 3: \frac {1}{7}WINNING PROBABILITY


ONE OF THESE WIN THE RACE:


=&gt; \frac {1}{7} +\frac {1}{8}+\frac {1}{7}


=&gt; \frac {8}{56} +\frac {7}{56} + \frac {8}{56} (TAKING LCM)


=&gt; \frac {23}{56}
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
There are 19 red balls and one black ball. Ten balls are put in one jar and the remaining 10 are put in another jar. What is the possibility that the black is in the right jar.

Options:
- **(A)** `1`
- **(B)** `1/2`
- **(C)** `2/3`
- **(D)** `1/4`

Correct Answer: (B) `1/2`

PrepInsta Detailed Explanation & Mathematical Derivation:
[Step-by-step logic directly follows from above axioms]

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:
There are 1000 junior and 800 senior students in a class. And there are 60 sibling pairs where each pair has 1 junior and 1 senior.1 student is chosen from senior and 1 from junior randomly. What is the probability that the two selected students are from a sibling pair?

Options:
- **(A)** `7140/800000`
- **(B)** `8450/800000`
- **(C)** `7455/800000`
- **(D)** `8230/800000`

Correct Answer: (A) `7140/800000`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Number of junior students: 1000


Number of Senior Students: 800


60 sibling pair = 2 x 60 = 120 students


The probability of the student being chosen from 800 and 1000 students is 1.


1 student chosen from senior and 1 student chosen from junior


n(s) = 800 x 1000 = 800000


n(E) = 120C2 = 7140


P(E) = n(E)/n(S) = 7140/800000
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
A bag contains 5 red caps, 4 blue caps, 3 yellow caps and 2 green caps. If four caps are picked at random, what is the probability that 2 are red, 1 blue and 1 green?

Options:
- **(A)** `80/1001`
- **(B)** `70/1001`
- **(C)** `60/1001`
- **(D)** `50/1001`

Correct Answer: (A) `80/1001`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Total no of caps= 5+4+3+2= 14


Probability= 5c2 4c1 2c1 / 14c4


80/1001
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
Two dice are thrown simultaneously. What is the probability that the sum of the numbers shown on the two

dices will be a prime number?

Options:
- **(A)** `17/36`
- **(B)** `1/2`
- **(C)** `15/36`
- **(D)** `7/18`

Correct Answer: (C) `15/36`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
number of sample solution=36


number of possible outcomes=(1,1),(1,2),(1,4),(1,6),(2,1),(2,3),(2,5),


(3,2)(3,4)(4,1)(4,3)(5,2)


(5,6)(6,1)(6,5)=15


p=15/36
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
There are 4 baskets. The first basket has 3 apples and 4 oranges, the second one has 4 apples and 5 mangoes, the third one has 6 Mangoes and 2 bananas and the last one has 7 bananas and 2 apples. If a fruit is randomly chosen from any basket and it comes out to be an apple, then what is the probability that it was taken out from the second basket?

Options:
- **(A)** `18/69`
- **(B)** `28/41`
- **(C)** `45/69`
- **(D)** `28/69`

Correct Answer: (D) `28/69`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
P(A|1)=\frac {3}{7}

P(A|2)=\frac {4}{9}

P(A|3)=\frac {0}{8}

P(A|4)=\frac {2}{9}

Hence event 1,2,3,4 are the partitions of the sample space.

The probability of choosing an apple, by the law of total probability

P(A)=P(A|1)P(1)+P(A|2)P(2)+P(A|3)P(3)+P(A|4)P(4)

=\frac {3}{7}x\frac {1}{4}+\frac {4}{9}x\frac {1}{4}+\frac {0}{8}x\frac {1}{4}+\frac {2}{9}x\frac {1}{4}

=\frac {23}{84}

By Baye\\\'s theorem,

P(2|A)=P(A|2)xP(2)/P(A)

=\frac{\frac {4}{9} \times \frac {1}{4}}{\frac {23}{84}}

=\frac {28}{69}
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
3 persons are standing at the middle of edges of a Triangle. All the 3 persons starts moving at same time with same speed in random direction What is the probability of meeting at least 2 persons?

Options:
- **(A)** `2/3`
- **(B)** `1`
- **(C)** `1/4`
- **(D)** `3/4`

Correct Answer: (D) `3/4`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
answer will be 3/4 as total no. of cases of possible moves=8

from this in 2 case their will not be collision(all anti clock or all clock wise)

so,p=(8-2)/8=3/4

No. of events will be 8.

This is because the person can move either inside or outside the triangle.

And the no. of favorable events will be 6. This is because we want

at least 2 person to meet and we have total 3 person, so 2x3 = 6

Probability = 6/8 = 3/4

Ans= 3/4
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:

There is a school were 60% are girls and 35% of the girls are poor. Students are selected at random, what is the probability of selecting a poor girl out of total strength?

Options:

- **(A)** `14/39`
- **(B)** `35/100`
- **(C)** `21/100`
- **(D)** `22/45`

Correct Answer: (C) `21/100`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let take 100 students

outoff 100 60 are girls

among girls poor girls are =35% 0f 60=21

pbt of selecting poor girl in total strength=21/100
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
What is the probability of drawn an ace or a space or both from a dew of cards.

Options:
- **(A)** `13/52`
- **(B)** `26/52`
- **(C)** `16/52`
- **(D)** `16/50`

Correct Answer: (C) `16/52`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
There are 13 spades in a standard deck of cards.

There are four aces in a standard deck of cards.

One of the aces is a spade.

So, 13 + 4 - 1 = 16 spades or aces to choose from.

Since we have a total of 52 cards, the probability of selecting an ace

or a spade is 16 / 52.
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

Options:

- **(A)** `1/4`
- **(B)** `1/3`
- **(C)** `1/2`
- **(D)** `2/3`

Correct Answer: (B) `1/3`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
ans = 1/3


becoz ther is only one chance to get tail on other side of coin,


amnog three coins,according to the


question
```
